

Rice Leaf Disease Classification using Deep Learning



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Abstract

Rice is one of the major crops in India, with Odisha as one of the major contributors to rice supply. With the increase in rice agriculture, several diseases can occur in the rice plants which is a concerning issue. Therefore, early detection of rice disease would play an important role, it will improve agriculture, food security, etc. This study aimed to build a neural network model to detect and classify rice leaf diseases. For this purpose, several papers were explored to get an overview of the topic and what steps could be taken for effective rice disease detection. Further, various actions were performed on the dataset to analyze the data, and by using the pre-trained model of Resnet-18 an accuracy of 95.47% was achieved.

Introduction

The objectives of this study were to explore numerous rice diseases including their symptoms and gain proficiency [1] in utilizing neural network algorithms and image processing techniques [2] to accurately detect and classify rice diseases from rice leaf images. The aim of this was to build a reliable model for early rice disease detection and find solutions to improve rice farming [3].

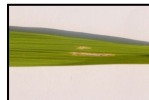


Figure 1. Bacterial Leaf Blight.

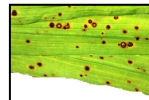


Figure 2. Brown Spot



Figure 3. Leaf Smut

Methodology

We selected the "Rice leaf diseases" dataset from Kaggle which consisted of 120 images, 40 from each class-bacterial, downloaded it and did the coding in VS code, and conducted thorough data cleaning and image augmentation which resulted in 40 images of each class to turn 690 for bacterial leaf blight, 585 for brown spot and 600 for leaf smut. The data was then split for training (80%) and testing (20%). Employing transfer learning models, including Alexnet, Resnet50 and Resnet18 we assessed their performance. Further, we used the Tkinter library to make the GUI for our app

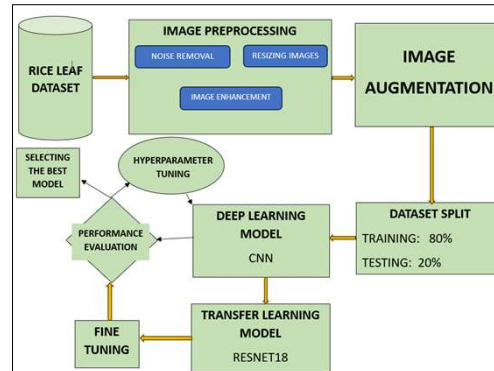


Figure 4. Methodology

Results & Discussion

The study on the "Rice leaf diseases" data showed that after sorting out the data and doing the required math, Resnet18 stood out as the best in predicting diseased leaves. It did well, with a validation accuracy of 95.47%. The way we looked at the data and used this method makes us pretty confident that it's a solid way to understand and predict leaf diseases in rice plants.

Model	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
Alexnet	0.2960	0.8598	0.2388	0.8966
Resnet50	0.3083	0.8552	0.2331	0.9034
Resnet18	0.2678	0.8853	0.1376	0.9547

Table 1. Best Performance of the models in 50 epochs



Figure 5. AlexNet: Epochs vs Loss & Epochs vs Accuracy

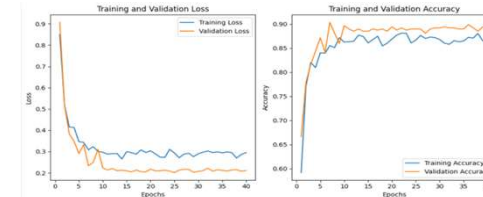


Figure 6. ResNet50: Epochs vs Loss & Epochs vs Accuracy

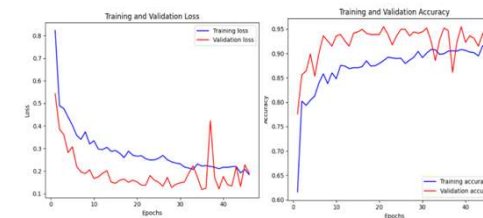


Figure 7. Resnet18: Epochs vs Loss & Epochs vs Accuracy

Conclusions

In conclusion, this project successfully developed an interface enabling users to upload rice leaf images for disease diagnosis. Through thorough data preprocessing and model selection, ResNet18 achieved a remarkable 95.47% accuracy in classifying bacterial leaf blight, brown spot, and leaf smut, facilitating effective and accessible rice disease detection.

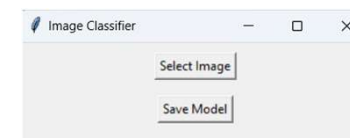


Figure 8. App interface

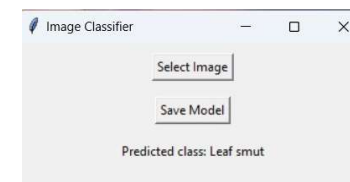


Figure 9. The model predicting the disease

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References

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- [2] S. Chen et al., "An approach for rice bacterial leaf streak disease segmentation and disease severity estimation," Agriculture (Switzerland), vol. 11, no. 5, 2021, doi: 10.3390/agriculture11050420.
- [3] H. Pallathadka et al., "Application of machine learning techniques in rice leaf disease detection," Mater Today Proc, vol. 51, pp. 2277–2280, 2022, doi: https://doi.org/10.1016/j.matpr.2021.11.398.

